

Physical Chemistry of Materials

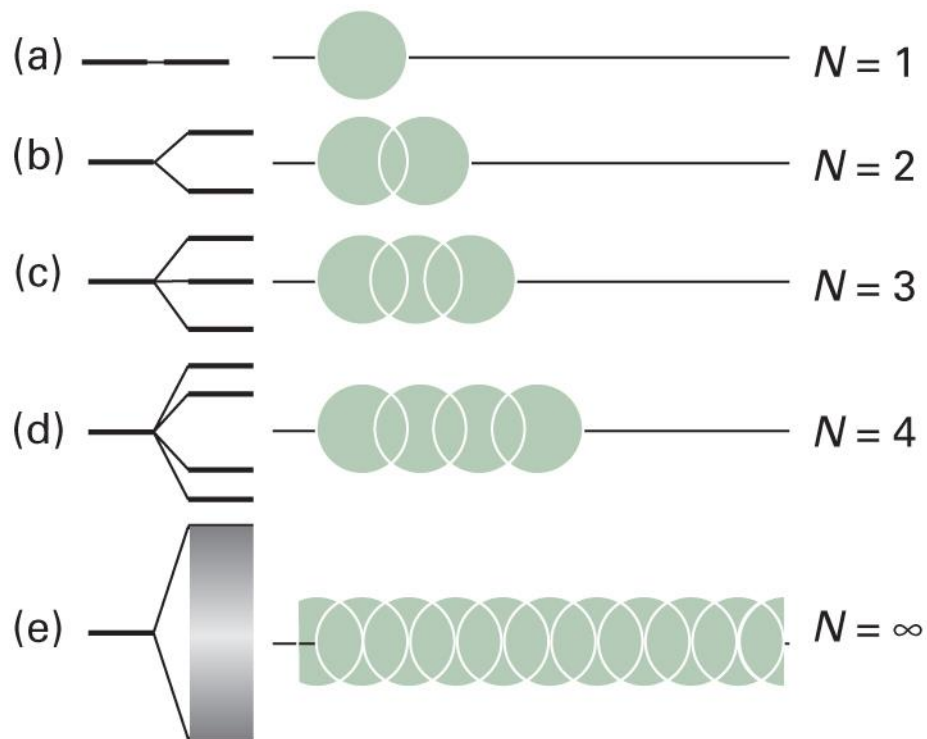
Lecture 14

Magnetism



Last Class

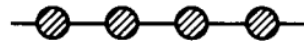
Electronic Structures of Solids



Bands

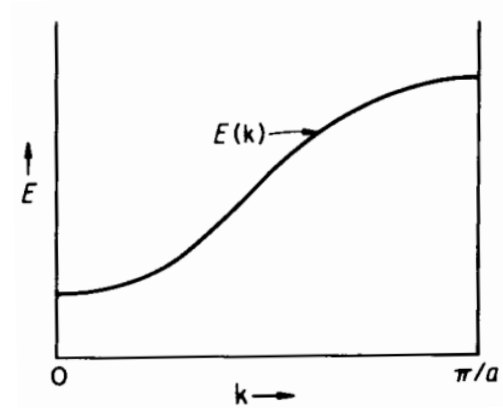
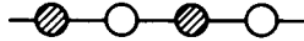
$$\psi_0 = \sum_n e^{i0n} \chi_n = \sum_n \chi_n$$

$$= \chi_0 + \chi_1 + \chi_2 + \chi_3 + \dots$$



$$\psi_{\frac{\pi}{a}} = \sum_n e^{i\pi n} \chi_n = \sum_n (-1)^n \chi_n$$

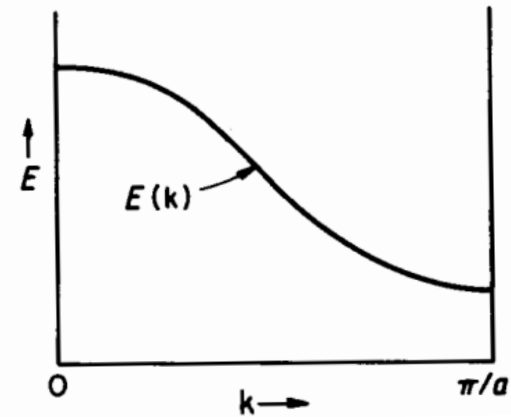
$$= \chi_0 - \chi_1 + \chi_2 - \chi_3 + \dots$$



$$\psi_0 = \chi_0 + \chi_1 + \chi_2 + \chi_3 + \dots$$

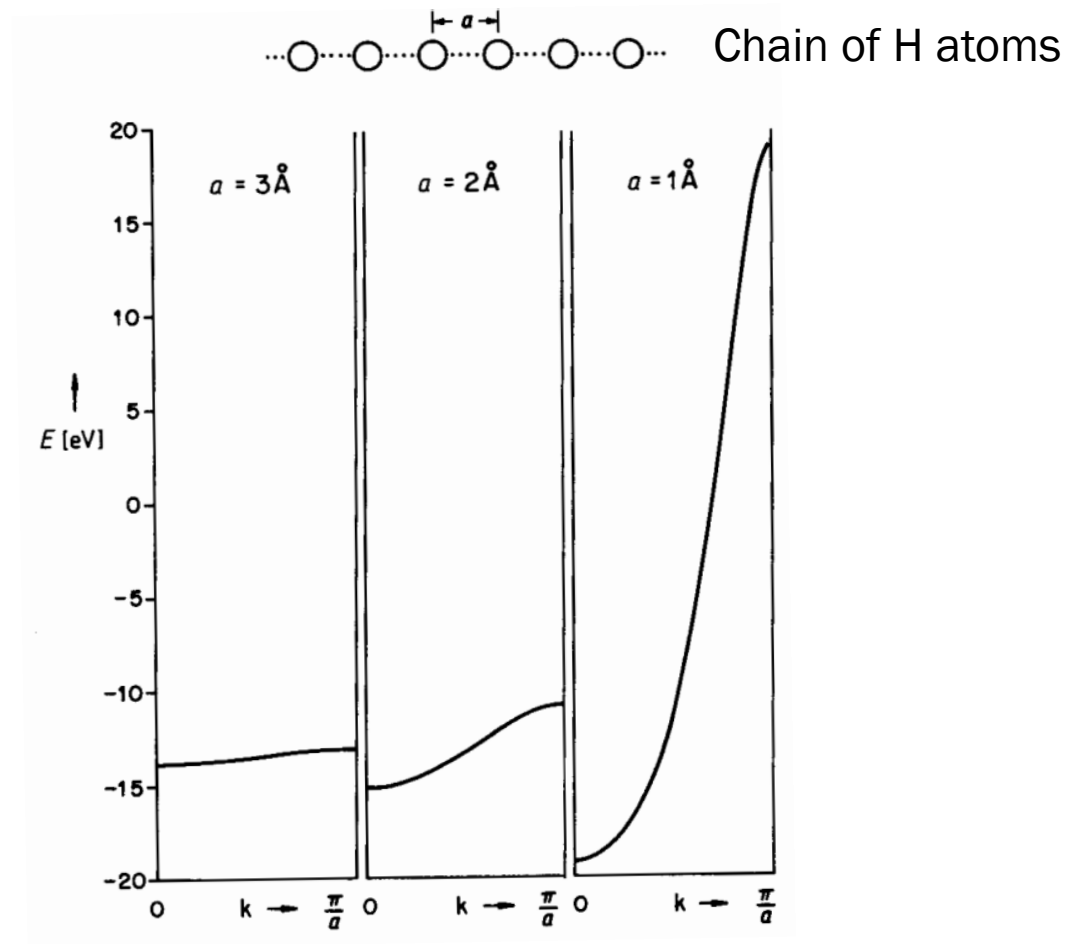


$$\psi_{\frac{\pi}{a}} = \chi_0 - \chi_1 + \chi_2 - \chi_3 + \dots$$



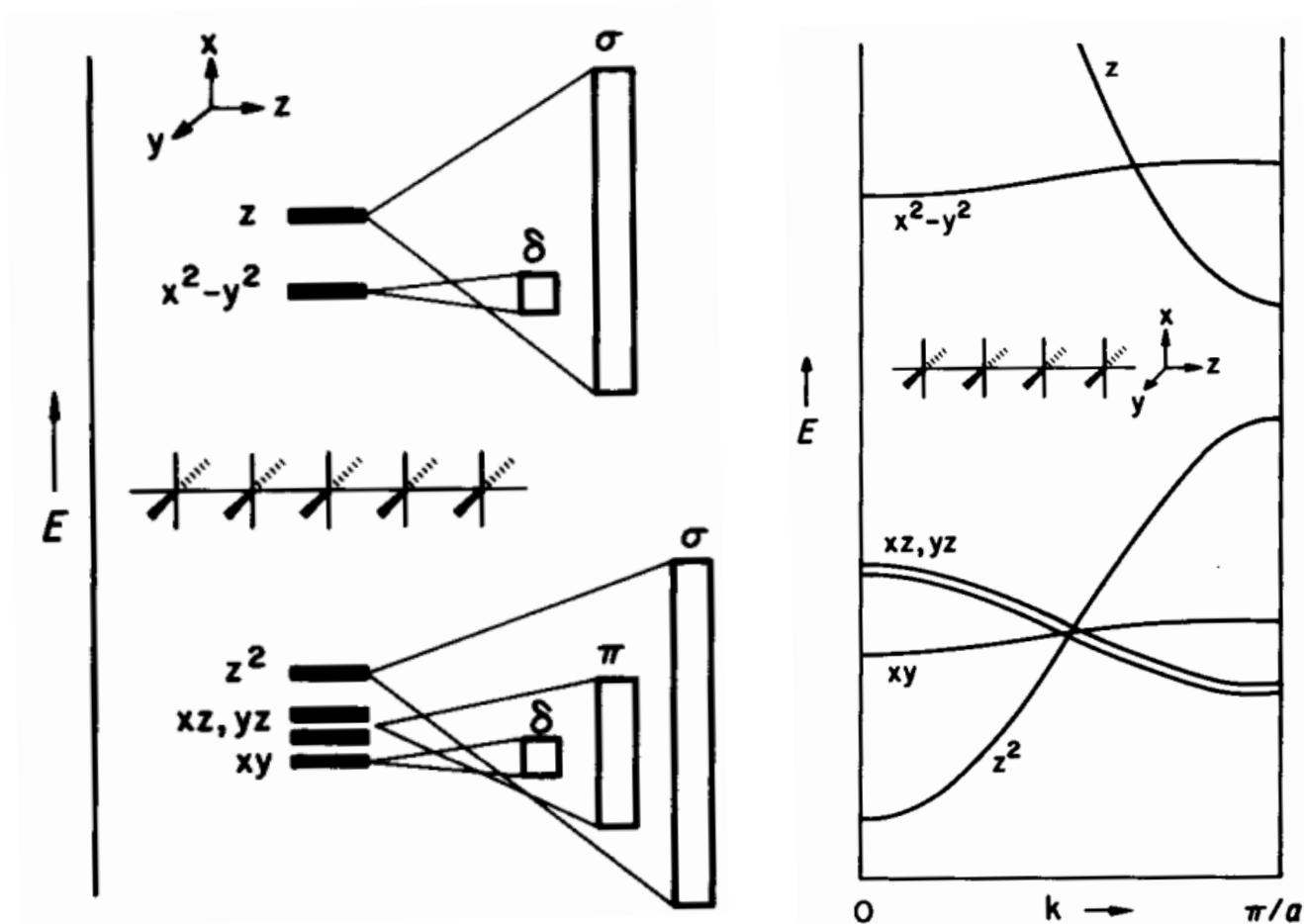
(Image: Hoffmann, *Solids and Surfaces*, p. 7 and p. 9)

“Band Width”



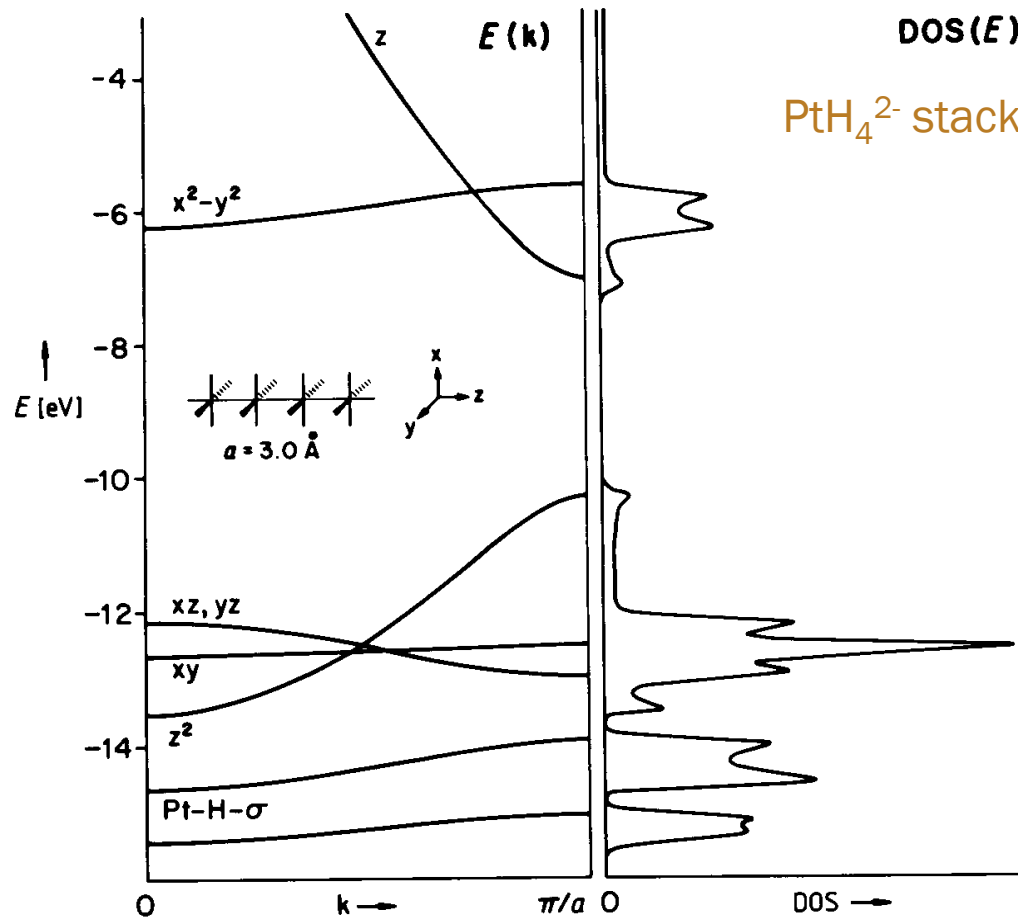
(Image: Hoffmann, *Solids and Surfaces*, Fig. 1)

Band Structure



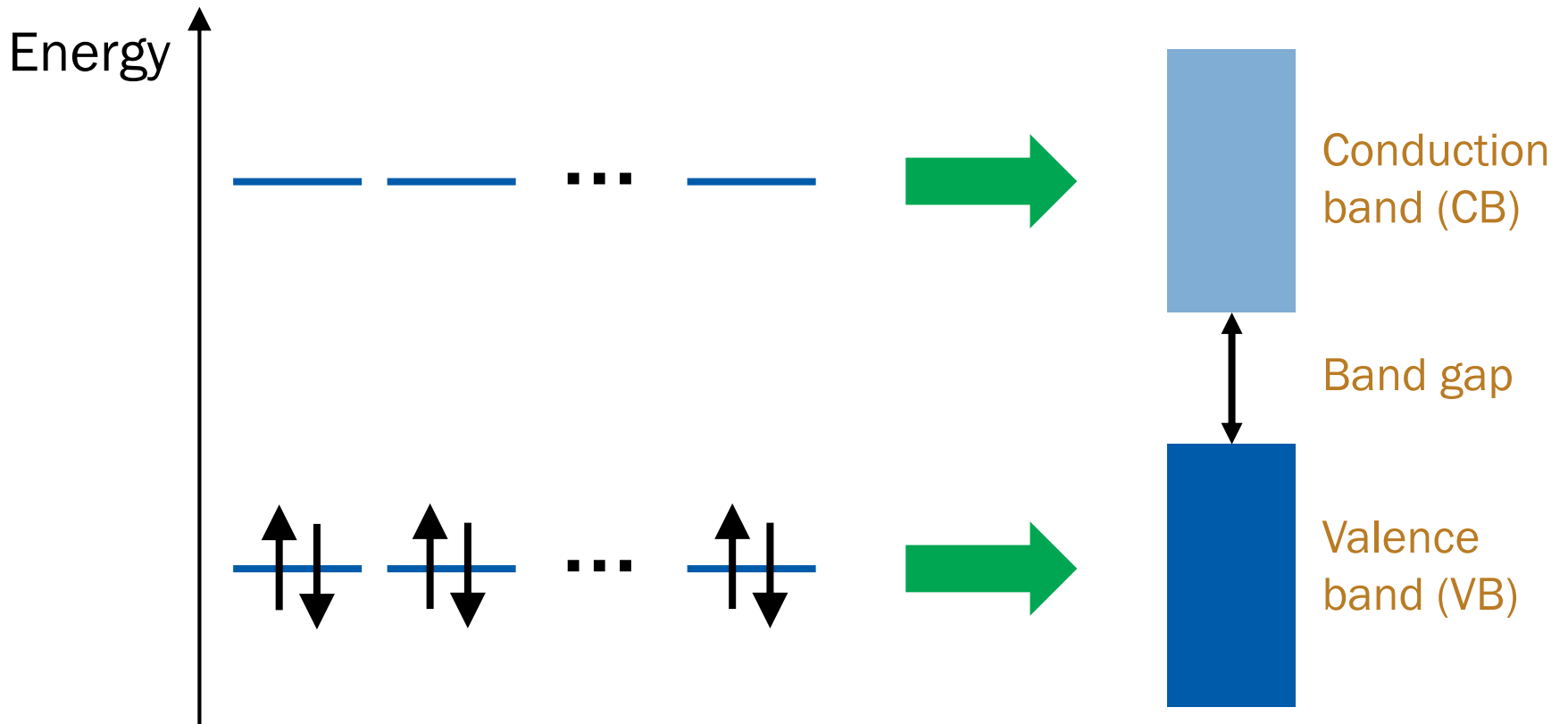
(Image: Hoffmann, *Solids and Surfaces*, p. 13)

DOS (Density of States)

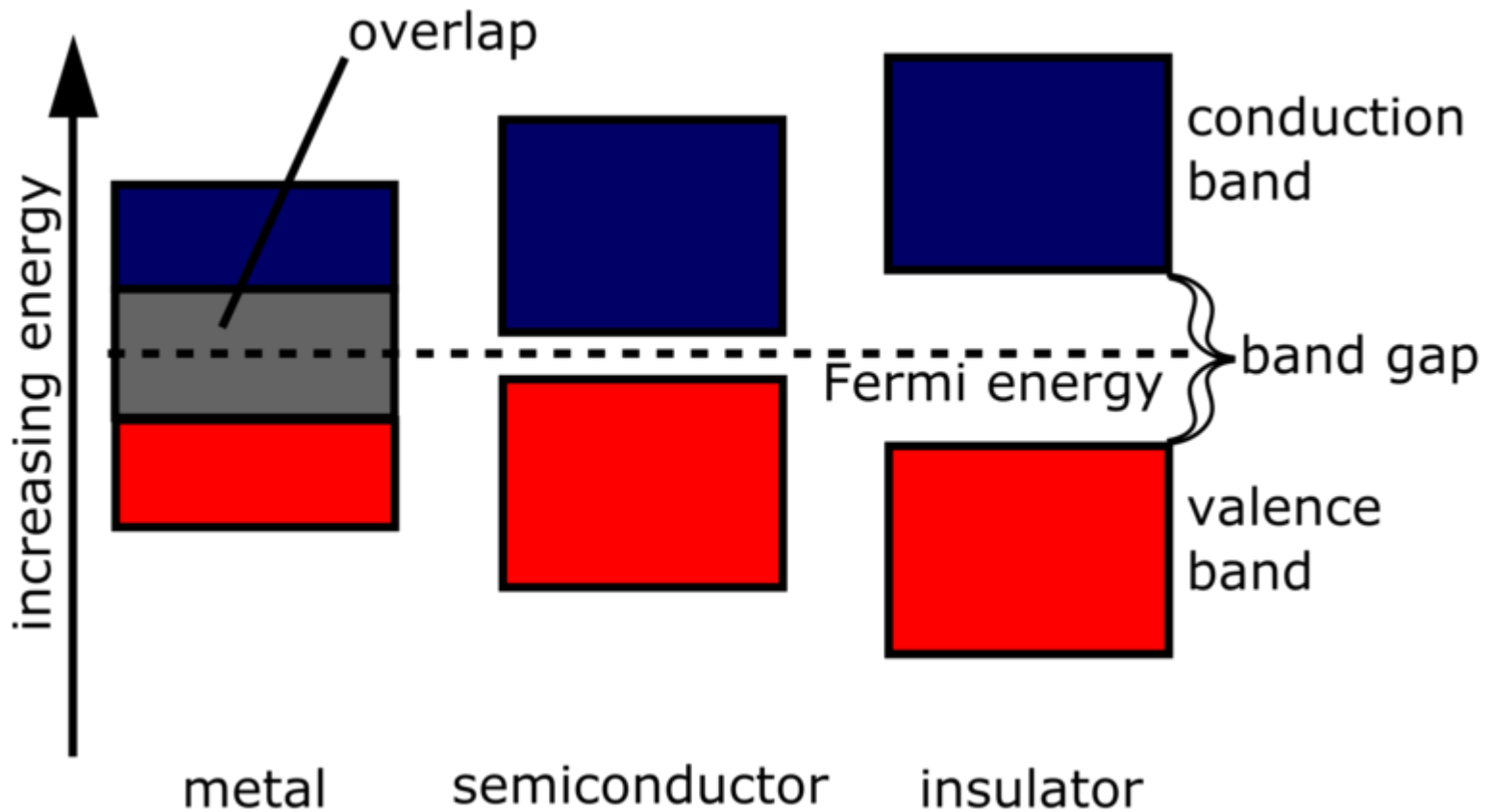


(Image: Hoffmann, *Solids and Surfaces*, p. 28)

Terminology

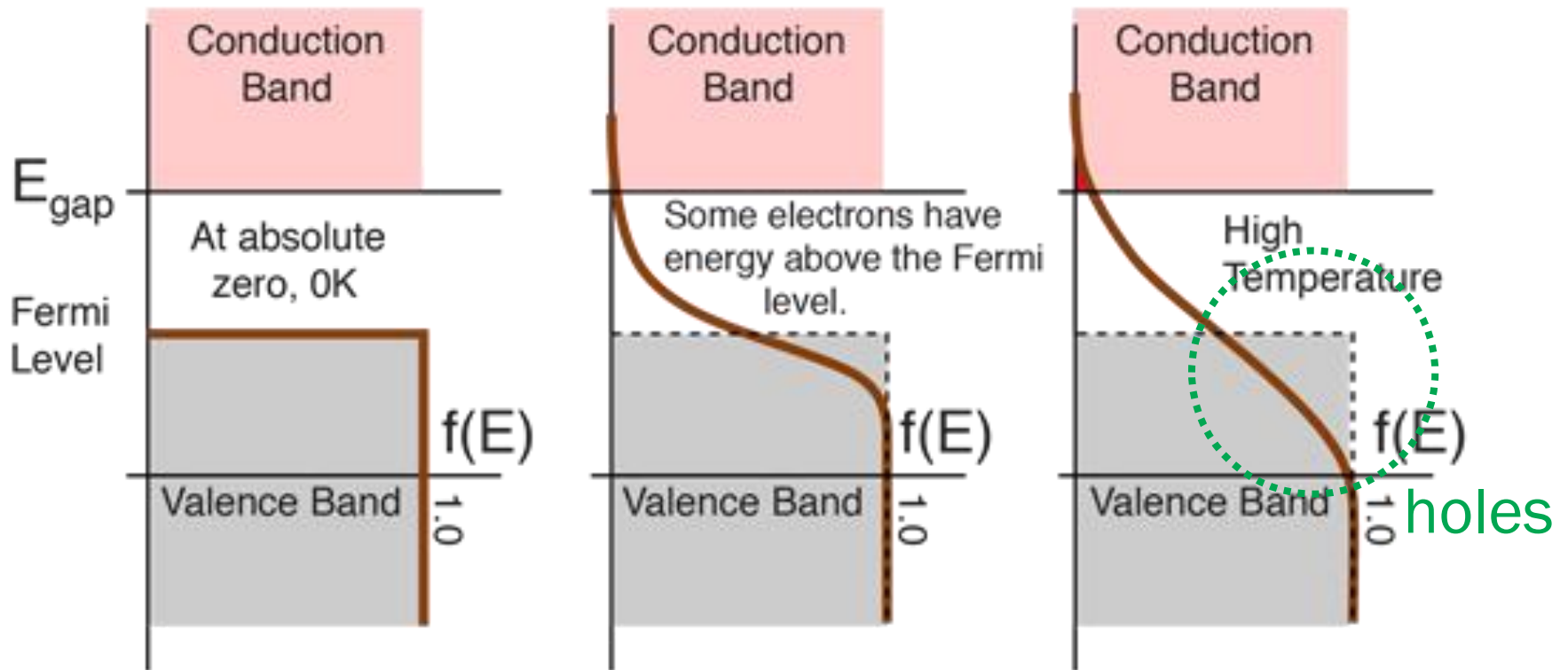


Band Gap and Conductivity



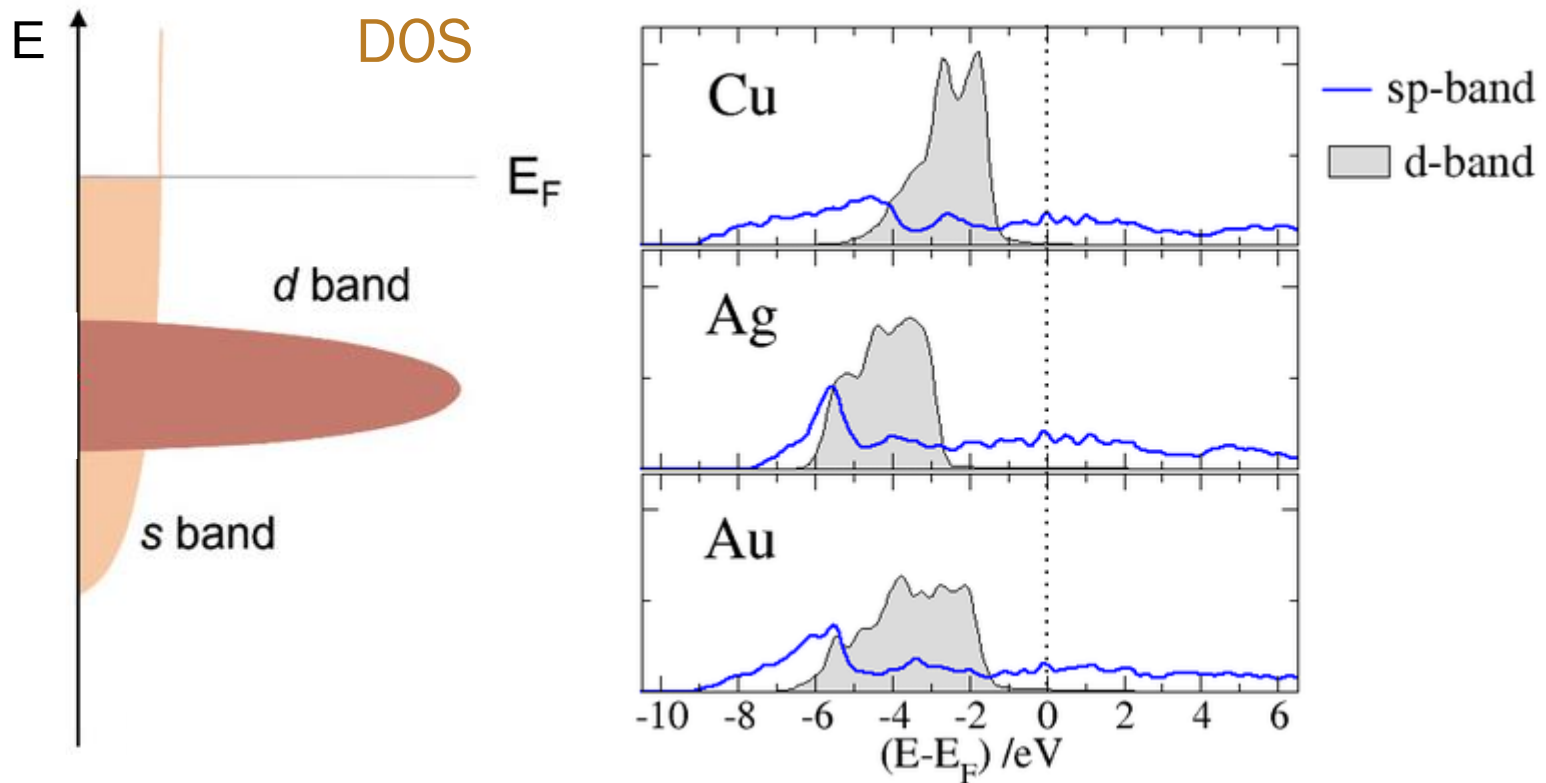
(Image: https://energyeducation.ca/encyclopedia/Band_gap)

At Finite Temperature



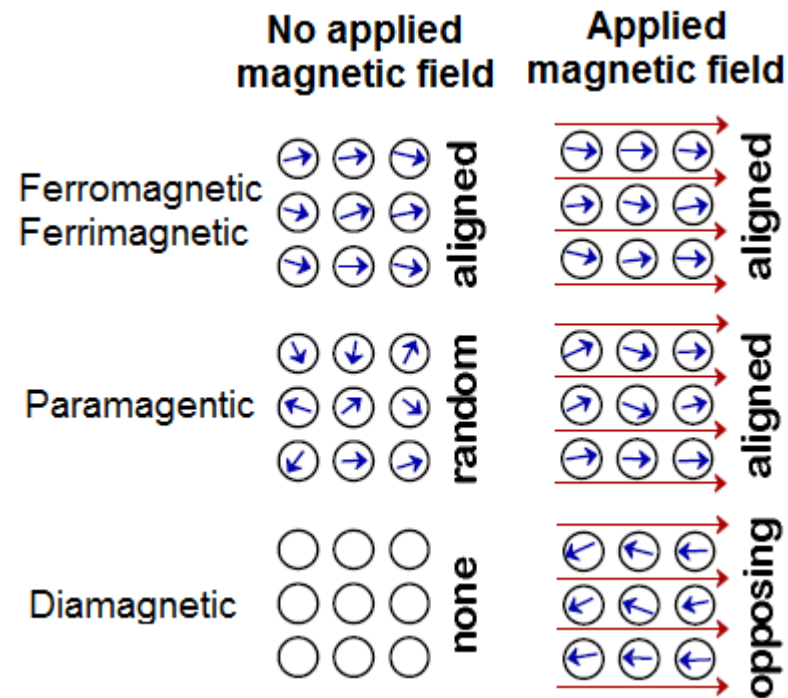
(Image: <http://hyperphysics.phy-astr.gsu.edu/hbase/Solids/Fermi.html>)

Colors of Cu, Ag, Au



(Images: Mittal et al., *Adv Mater Interfaces* (2022); <https://doi.org/10.13140/2.1.4142.3364>)

Magnetism



Origin of Magnetization

For a free electron in a magnetic field,

$$E_{\text{spin}} = g\mu_B m_s B$$

where

- g : g -factor ($g \approx 2$)
- μ_B : Bohr magneton ($\mu_B = e\hbar/2m_e$)
- m_s : spin magnetic quantum number, $m_s = \pm 1/2$
- B : amplitude of the magnetic field

Microstates

$$E_{\text{spin}} = g\mu_B m_s B \approx 2\mu_B m_s B$$

Consider two spin states, up and down:

$$E(\uparrow) = \mu_B B$$

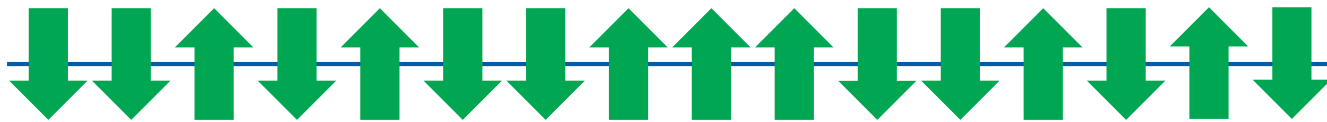
$$E(\downarrow) = -\mu_B B$$

With $\sigma = \pm 1$, we can also write

$$E(\sigma) = \mu_B B \sigma$$

Independent Spin Model

Consider a solid system consisting of N spins:



$$E(\uparrow) = +\mu_B B, \quad E(\downarrow) = -\mu_B B$$

The system consists of distinguishable particles:

$$Z = z^N$$

Partition Function

The one-particle partition function is

$$z = \sum_{\sigma} e^{-\beta E(\sigma)} = e^{-\beta E(\uparrow)} + e^{-\beta E(\downarrow)}$$

$$z = e^{-\beta \mu_B B} + e^{+\beta \mu_B B} = 2 \cosh(\beta \mu_B B)$$

Hence,

$$Z = z^N = [2 \cosh(\beta \mu_B B)]^N$$

Thermodynamics

- Helmholtz energy

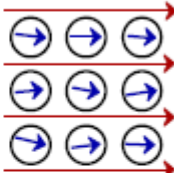
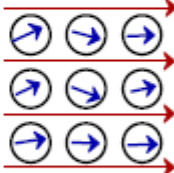
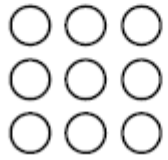
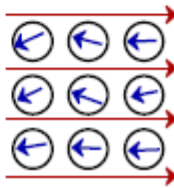
$$A = -k_B T \ln Z = -N k_B T \ln[2 \cosh(\beta \mu_B B)]$$

- Internal energy

$$U = -\frac{\partial \ln Z}{\partial \beta} = -N \frac{\partial}{\partial \beta} \ln[2 \cosh(\beta \mu_B B)]$$

$$\therefore U = -N \mu_B B \tanh(\beta \mu_B B)$$

Magnetism

	No applied magnetic field	Applied magnetic field	
 	Ferromagnetic Ferrimagnetic  aligned	 aligned	
	Paramagnetic  random	 aligned	$S \neq 0$ (or $J \neq 0$)
	Diamagnetic  none	 opposing	$S = 0$ (or $J = 0$)

Ising Model

Consideration of neighboring spin-spin interactions

- System energy

$$E = \mu_B B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

- Sign of J
 - $J > 0$: ferromagnetic
 - $J < 0$: antiferromagnetic

Partition Function of Ising Model

$$E = \mu_B B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

$$Z = \sum_{\text{microstate}} e^{-\beta E} = \sum_{\{\sigma_i\}} e^{-\beta E(\{\sigma_i\})}$$

$$\sum_{\{\sigma_i\}} = \sum_{\sigma_1=\pm 1} \sum_{\sigma_2=\pm 1} \sum_{\sigma_3=\pm 1} \cdots \sum_{\sigma_N=\pm 1}$$

Ising Model: Solved?

depends on the dimensionality D

- $D = 1$: exactly solvable for all J and $\mu_B B$
- $D = 2$: exactly solvable for $\mu_B B = 0$
- $D = 3$: not yet solved
- $D \geq 4$: solvable by the mean-field approach

Mean-Field Approach

$$E = \mu_B B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

For spin i ,

$$E_i = \mu_B B \sigma_i - J \sigma_i \sum_{\text{n.n. } j} \sigma_j \approx \mu_B B \sigma_i - J \sigma_i \sum_{\text{n.n. } j} \langle \sigma_j \rangle$$

By denoting $\langle \sigma_j \rangle = m$, for the coordination number z ,

$$E_i = \mu_B B \sigma_i - J \sigma_i z m$$

Mean-Field Approach

$$E_i = \mu_B B \sigma_i - J \sigma_i z m = (\mu_B B - J z m) \sigma_i$$

On the other hand,

$$m = \langle \sigma_i \rangle = \sum_{\sigma_i = \pm 1} \sigma_i p_i = \sum_{\sigma_i} \frac{\sigma_i e^{-\beta E_i}}{Z} = \frac{\sum_{\sigma_i} \sigma_i e^{-\beta E_i}}{\sum_{\sigma_i} e^{-\beta E_i}}$$

Thus,

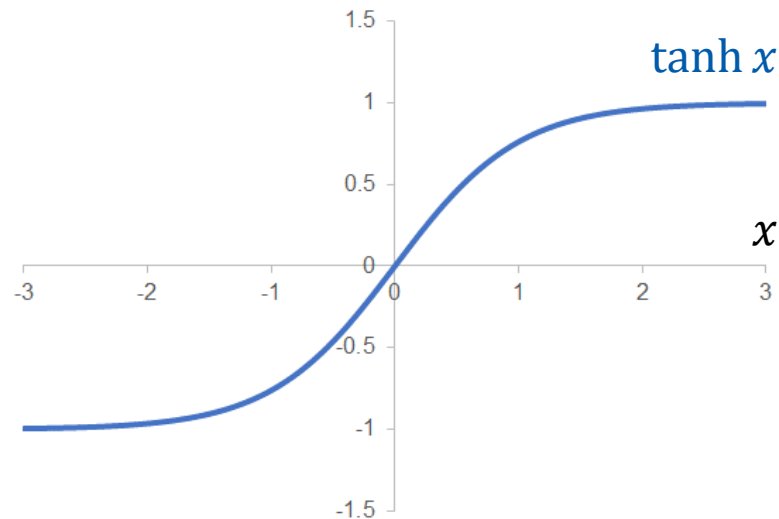
$$m = \frac{e^{-\beta(\mu_B B - J z m)} - e^{\beta(\mu_B B - J z m)}}{e^{-\beta(\mu_B B - J z m)} + e^{\beta(\mu_B B - J z m)}} = \tanh[\beta(J z m - \mu_B B)]$$

(self-consistent equation)

Graphical Solution

$$m = \tanh[\beta(Jzm - \mu_B B)]$$

For $B = 0$, by letting $x = \beta Jzm$, we obtain $\frac{k_B T}{Jz} x = \tanh x$



Phase Transition

- High T (steep slope): one solution at $x = 0$
- Low T (shallow slope): three solutions
- Critical temperature T_c

$$\frac{k_B T}{Jz} x = \tanh x$$

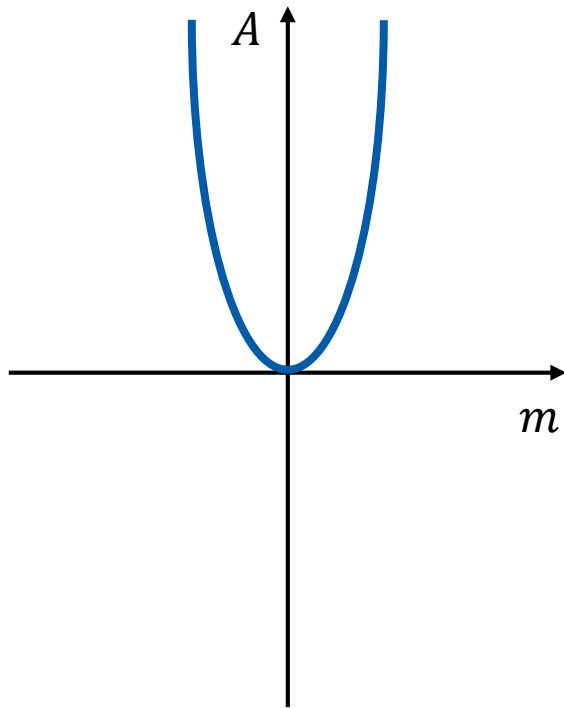
Since $\lim_{x \rightarrow 0} \tanh x = x$,

$$k_B T_c = Jz$$

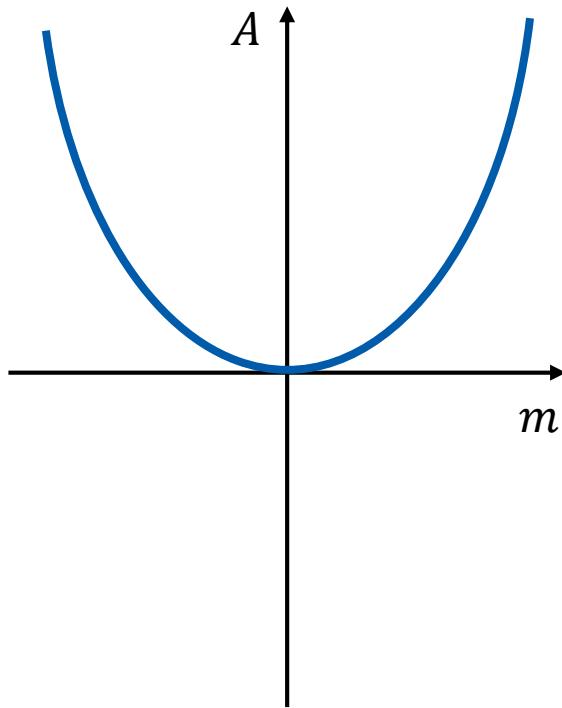
$$\therefore T_c = \frac{Jz}{k_B}$$

What Happens?

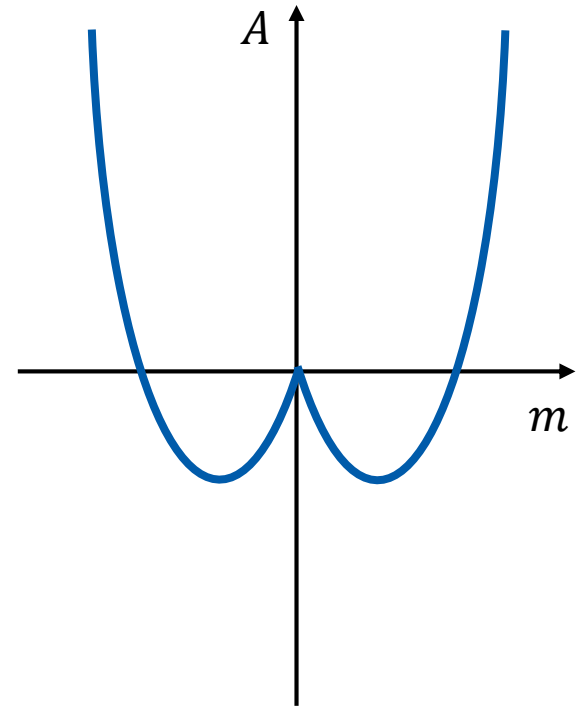
$$T > T_c$$



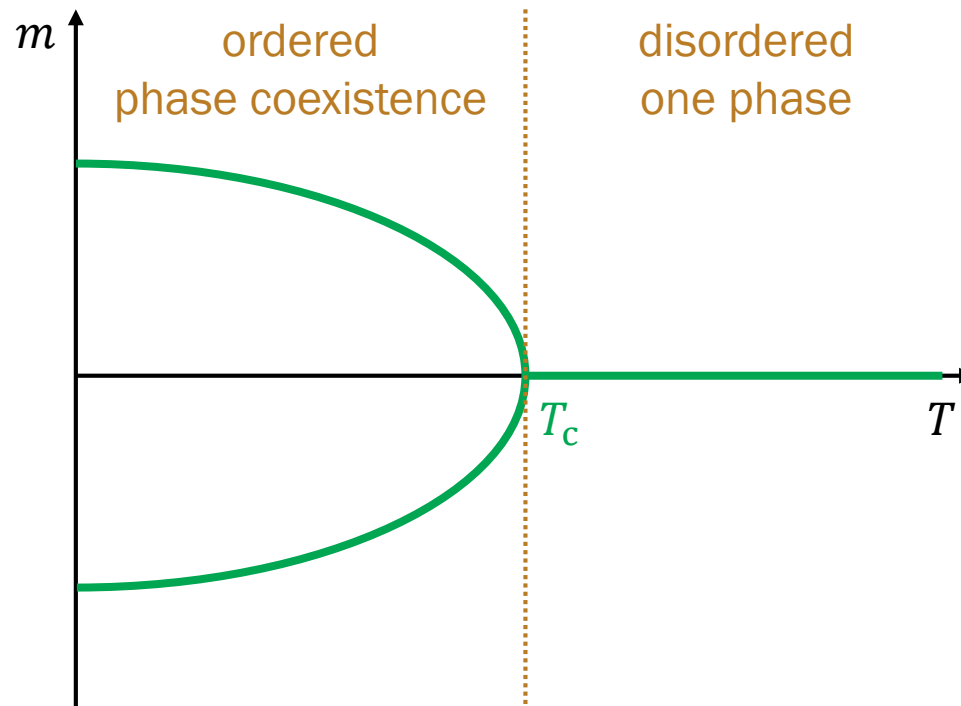
$$T = T_c$$



$$T < T_c$$

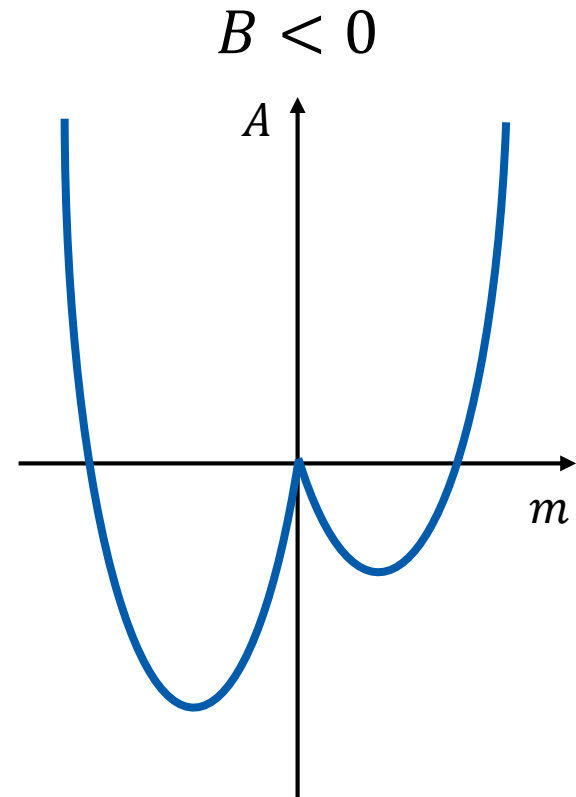
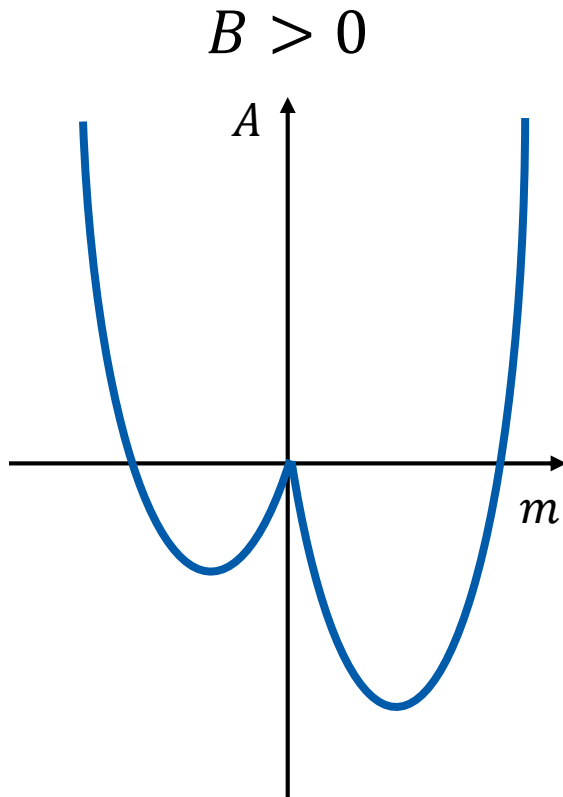


Positions of Minima

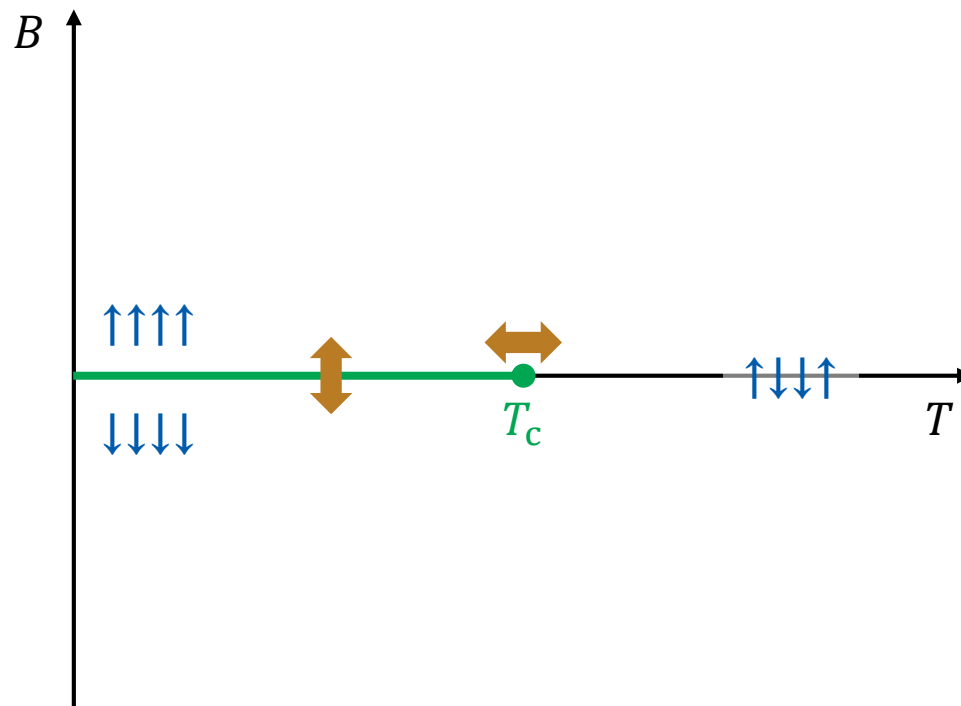


What If $B \neq 0$?

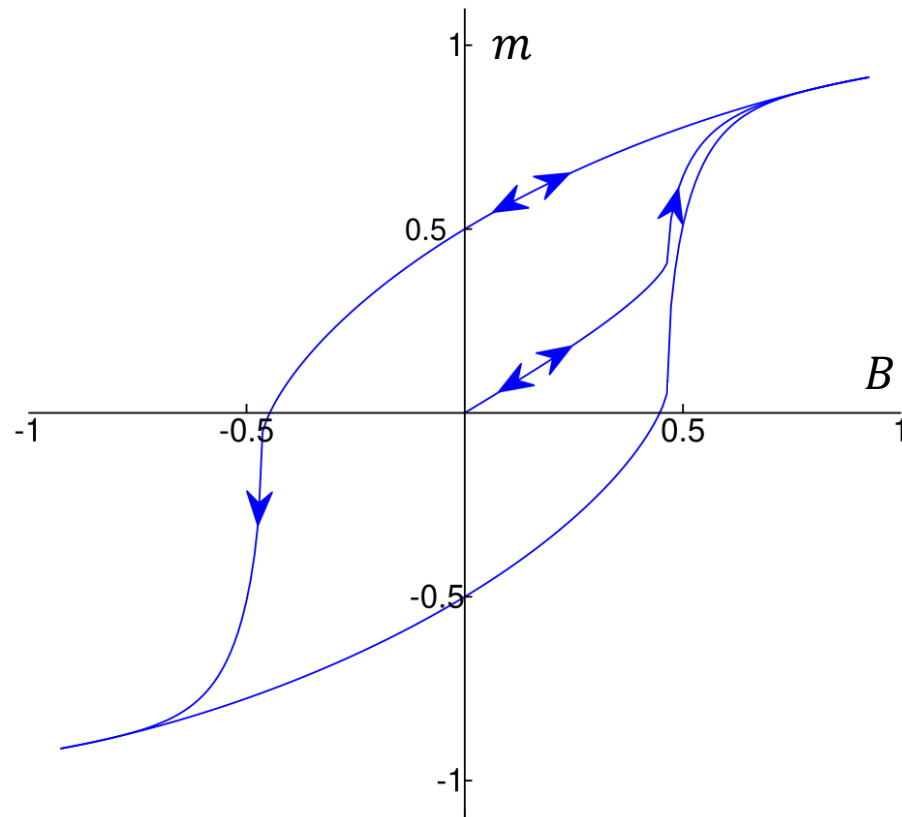
$(T < T_c)$



Phase Diagram



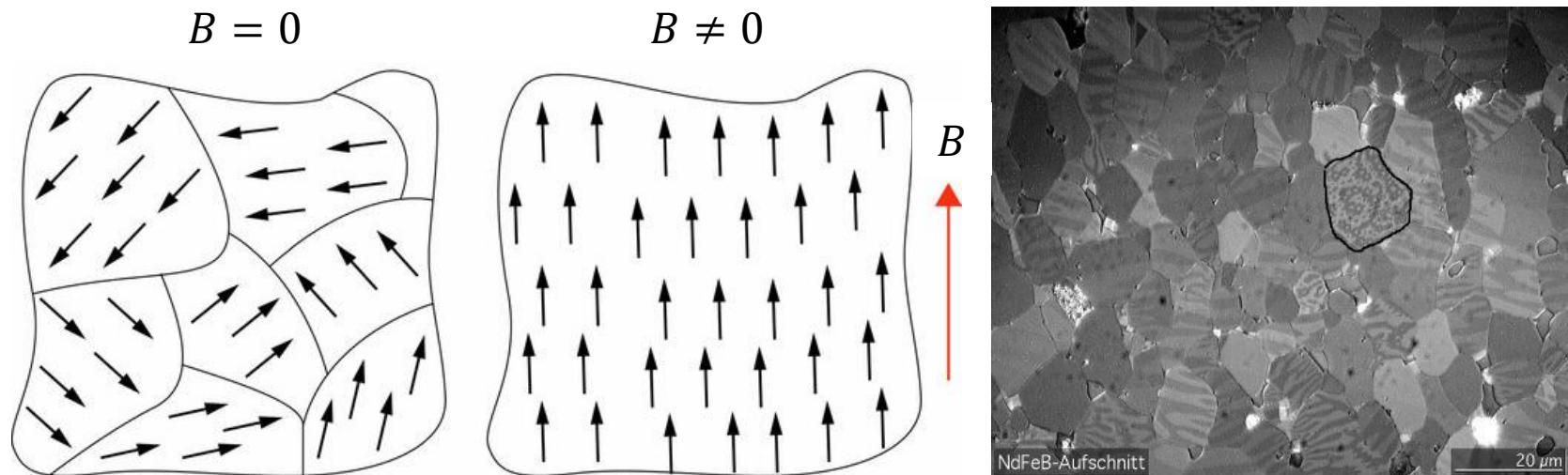
Magnetic Hysteresis



(Image: <https://commons.wikimedia.org/wiki/File:StonerWohlfarthMainLoop.svg>)

Magnetic Domains

Why a ferromagnet not magnetic when $B = 0$?



1-Dimensional Ising Model

For $B = 0$,

$$Z = \sum_{\{\sigma_i\}} e^{\beta J \sum_{\langle ij \rangle} \sigma_i \sigma_j}$$

Consider the periodic boundary condition ($\sigma_{N+1} = \sigma_1$);

$$Z = \sum_{\sigma_1=\pm 1} \sum_{\sigma_2=\pm 1} \sum_{\sigma_3=\pm 1} \dots \sum_{\sigma_N=\pm 1} e^{\beta J \sum_i \sigma_i \sigma_{i+1}}$$

1-Dimensional Ising Model

$$Z = \sum_{\sigma_1=\pm 1} \sum_{\sigma_2=\pm 1} \sum_{\sigma_3=\pm 1} \cdots \sum_{\sigma_N=\pm 1} e^{\beta J \sum_i \sigma_i \sigma_{i+1}}$$

Define $\tau_i = \sigma_i \sigma_{i+1} = \pm 1$;

$$Z = \sum_{\tau_1=\pm 1} \sum_{\tau_2=\pm 1} \sum_{\tau_3=\pm 1} \cdots \sum_{\tau_N=\pm 1} e^{\beta J \sum_i \tau_i}$$

$$Z = \left(\sum_{\tau=\pm 1} e^{\beta J \tau} \right)^N = [2 \cosh(\beta J)]^N$$

Thermodynamics

- Partition function $Z = [2 \cosh(\beta J)]^N$
- Helmholtz energy $A = -Nk_B T \ln[2 \cosh(\beta J)]$
- Energy $U = -NJ \tanh(\beta J)$

$$\lim_{T \rightarrow 0} U = -NJ$$

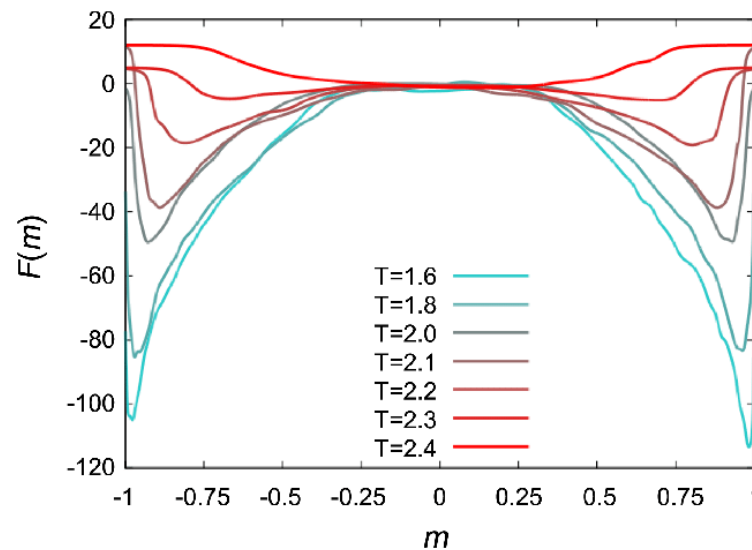
$$\lim_{T \rightarrow \infty} U = 0$$

No phase transition!

2D Ising Model

- Partition function (Onsager 1944)

$$\frac{\ln Z}{N} = \ln 2 + \frac{1}{8\pi^2} \int_{-\pi}^{\pi} \ln[\cosh^2(2K) - \sinh(2K)(\cos q_x + \cos q_y)] dq_x dq_y$$



(Image: Oh et al., *Bull Kor Chem Soc* 2012)

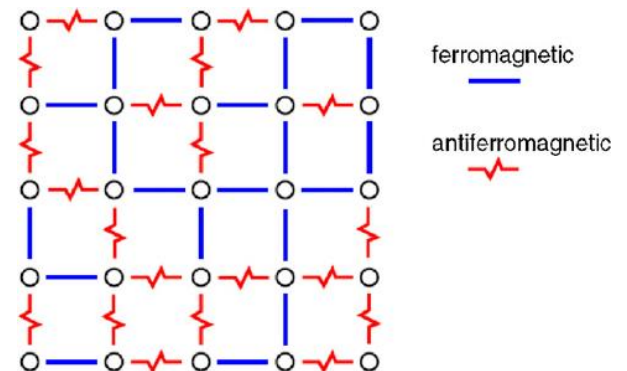
Spin Glasses

- Ising model

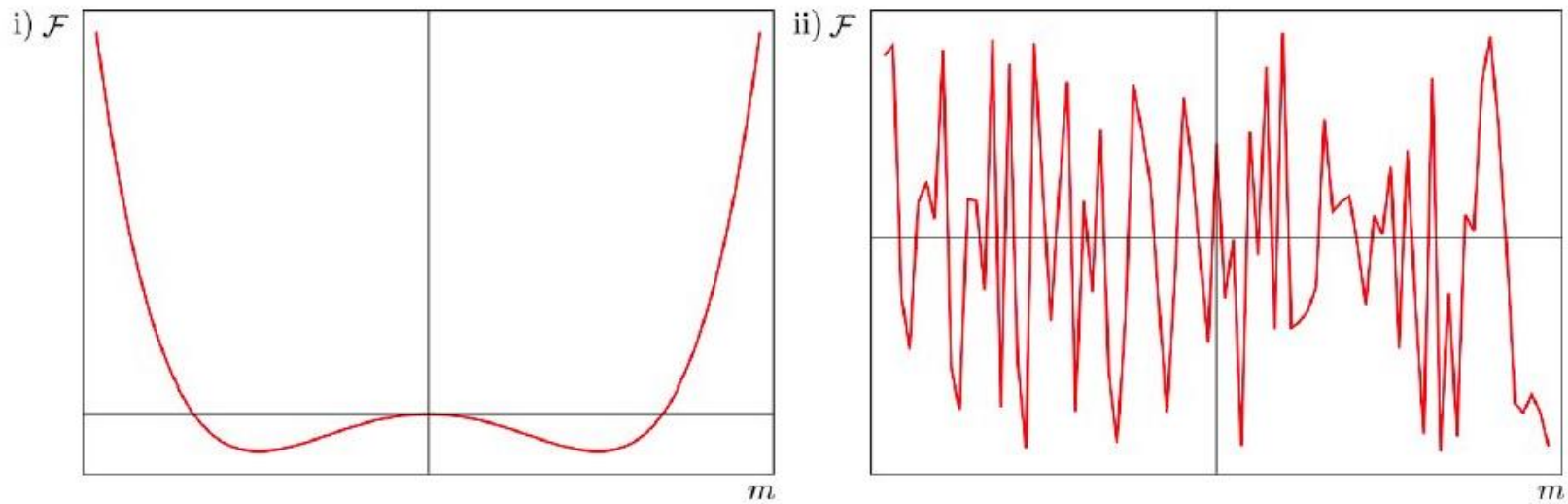
$$E = \mu_B B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

- Edwards-Anderson model

$$E = - \sum_{\langle ij \rangle} J_{ij} \sigma_i \sigma_j$$



Disordered Systems



Nobel prize in Physics 2021

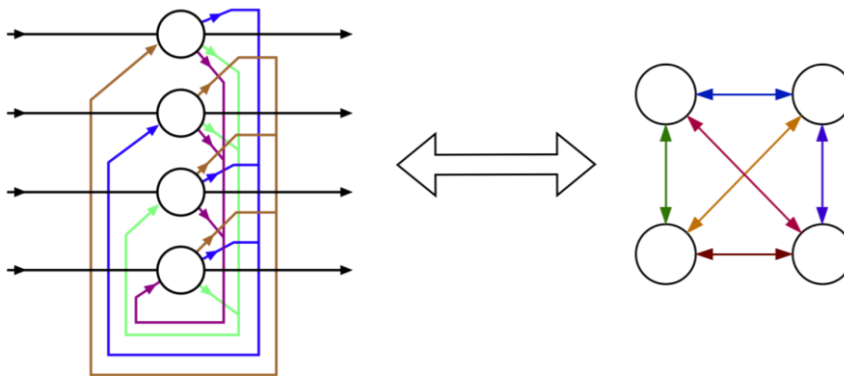
Hopfield Model

- Time evolution

$$\sigma_i(t + \Delta t) = \text{sgn}\left(\sum_j J_{ij}\sigma_j(t)\right)$$

- Equivalent energy

$$E = - \sum_{\langle ij \rangle} J_{ij} \sigma_i \sigma_j$$



Nobel prize in
Physics 2024

(Image: <https://www.zach.be/p/implementing-hopfield-networks-on>)